



Programming Fundamentals

Selection 2 — Multiple Choices and Logical Operators

Programming Fundamentals Team

SETU

2026-09-15



Outline

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1. Today

By the end of this session you should be able to:

- Use `if-elif-else` for multiple alternatives.
- Explain the difference between separate `ifs` and an `elif` chain.
- Combine conditions using `and`, `or` and `not`.
- Check ranges correctly.
- Use nested selection when appropriate.
- Apply selection to validation and small programs.



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2. Quick Recap

What will this print?

```
mark = 40
```

```
if mark >= 40:  
    print("Pass")  
else:  
    print("Fail")
```



2. Quick Recap

What will this print?

```
mark = 40
```

```
if mark >= 40:  
    print("Pass")  
else:  
    print("Fail")
```

Pass — because 40 is included in `>= 40`.



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3. More Than Two Choices

Suppose we want:

- 70+ $\rightarrow$ Distinction
- 40–69 $\rightarrow$ Pass
- below 40 $\rightarrow$ Fail



3. More Than Two Choices

Suppose we want:

- 70+ → Distinction
- 40–69 → Pass
- below 40 → Fail

An `if-else` only gives us two branches.

We need `elif` — **else if**.



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4. if-elif-else Syntax

```
if condition1:
    statements
elif condition2:
    statements
else:
    statements
```



4. if-elif-else Syntax

```
if condition1:  
    statements  
elif condition2:  
    statements  
else:  
    statements
```

Python tests from top to bottom and stops at the **first true branch**.



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5. Grade Classification

```
mark = int(input("Mark: "))

if mark >= 70:
    print("Distinction")
elif mark >= 40:
    print("Pass")
else:
    print("Fail")
```



5. Grade Classification

```
mark = int(input("Mark: "))

if mark >= 70:
    print("Distinction")
elif mark >= 40:
    print("Pass")
else:
    print("Fail")
```

Why does the second condition not need `mark < 70`?



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6. Order Matters

This is wrong:

```
if mark >= 40:
    print("Pass")
elif mark >= 70:
    print("Distinction")
else:
    print("Fail")
```



6. Order Matters

This is wrong:

```
if mark >= 40:
    print("Pass")
elif mark >= 70:
    print("Distinction")
else:
    print("Fail")
```

For `mark = 85`, the first condition is already true.

The `elif` is never reached.

When testing thresholds, usually test the most restrictive/highest threshold first.



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7. Trace the Chain

For each value, which branch runs?

```
if score >= 80:  
    print("A")  
elif score >= 60:  
    print("B")  
elif score >= 40:  
    print("C")  
else:  
    print("D")
```

- 90
- 65
- 40
- 39



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8. Separate ifs or elif?

```
if temperature > 20:  
    print("Warm")  
if temperature > 30:  
    print("Very warm")
```



8. Separate ifs or elif?

```
if temperature > 20:  
    print("Warm")  
if temperature > 30:  
    print("Very warm")
```

If temperature is 35, **both** messages print.



8. Separate ifs or elif?

```
if temperature > 20:  
    print("Warm")  
if temperature > 30:  
    print("Very warm")
```

If temperature is 35, **both** messages print.

With `elif`, only one branch in the chain can run.



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9. Choosing the Right Structure

Use separate `if` statements when:

- several conditions may all be true.

Use `if-elif-else` when:

- the choices are mutually exclusive.



9. Choosing the Right Structure

Use separate `if` statements when:

- several conditions may all be true.

Use `if-elif-else` when:

- the choices are mutually exclusive.

Example: a person can be both a student **and** employed — separate tests may make sense.



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10. Logical Operators

Sometimes one comparison is not enough.

Python provides:

- and — both conditions must be true
- or — at least one condition must be true
- not — reverses a Boolean value



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11. and

A theme park ride requires:

- age 12 or older
- height at least 140 cm

```
if age >= 12 and height >= 140:  
    print("You may ride")  
else:  
    print("Sorry")
```

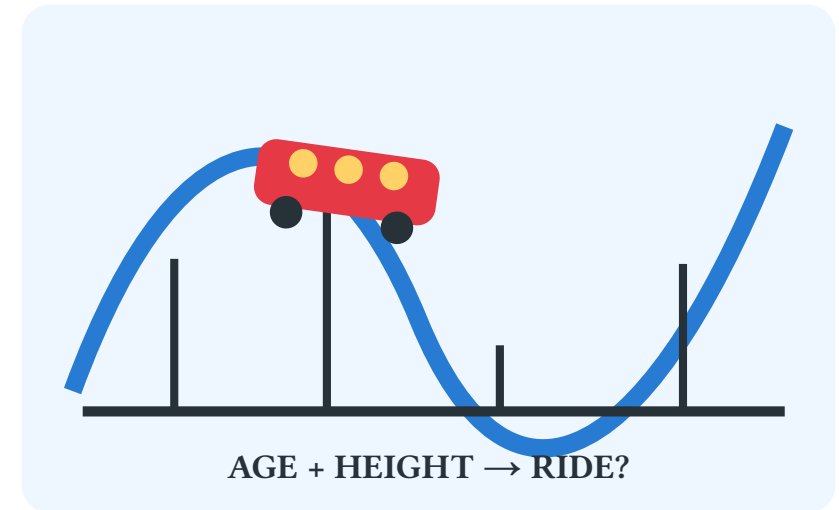
11. and

A theme park ride requires:

- age 12 or older
- height at least 140 cm

```
if age >= 12 and height >= 140:  
    print("You may ride")  
else:  
    print("Sorry")
```

Both tests must be True.





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12. Truth Table: and

A	B	A and B
True	True	True
True	False	False
False	True	False
False	False	False



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13. or

A discount is available if the customer is:

- a student, **or**
- aged 65 or over.

```
if is_student or age >= 65:  
    print("Discount applies")
```



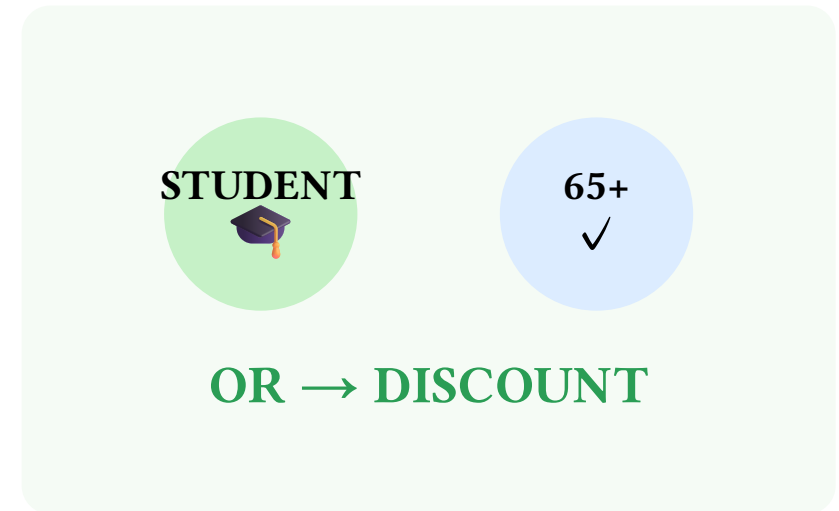
13. or

A discount is available if the customer is:

- a student, **or**
- aged 65 or over.

```
if is_student or age >= 65:  
    print("Discount applies")
```

Only one condition needs to be True.





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14. Truth Table: or

A	B	A or B
True	True	True
True	False	True
False	True	True
False	False	False



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15. not

not reverses a Boolean result.

```
logged_in = False
```

```
if not logged_in:  
    print("Please log in")
```



15. not

not reverses a Boolean result.

```
logged_in = False
```

```
if not logged_in:  
    print("Please log in")
```

not False becomes True.



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16. Checking a Range

Is mark between 40 and 69 inclusive?

Python allows chained comparisons:

```
if 40 <= mark <= 69:  
    print("Pass")
```



16. Checking a Range

Is mark between 40 and 69 inclusive?

Python allows chained comparisons:

```
if 40 <= mark <= 69:  
    print("Pass")
```

You can also write:

```
if mark >= 40 and mark <= 69:
```

The first form is especially readable in Python.



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17. Common Range Error

This does **not** mean between 40 and 69:

```
if mark >= 40 or mark <= 69:
```



17. Common Range Error

This does **not** mean between 40 and 69:

```
if mark >= 40 or mark <= 69:
```

Why?

Almost every number satisfies at least one side.

For a value to be inside a range, we usually need and.



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18. Activity: Login Rule

A user may log in only when:

- username is student
- password is python123

Write the condition.

```
username = input("Username: ")  
password = input("Password: ")
```





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19. Login Solution

```
if username == "student" and password == "python123":  
    print("Access granted")  
else:  
    print("Access denied")
```



19. Login Solution

```
if username == "student" and password == "python123":  
    print("Access granted")  
else:  
    print("Access denied")
```

Extension: should passwords be case-sensitive? Should usernames?



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20. Nested Selection

An if can appear inside another if.

```
if age >= 18:
    if has_ticket:
        print("Enter")
    else:
        print("You need a ticket")
else:
    print("You must be 18+")
```



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21. Nested or and?

Sometimes nesting explains **why** a user failed:

```
if age >= 18:
    if has_ticket:
        print("Enter")
    else:
        print("You need a ticket")
else:
    print("You must be 18+")
```



21. Nested or and?

Sometimes nesting explains **why** a user failed:

```
if age >= 18:
    if has_ticket:
        print("Enter")
    else:
        print("You need a ticket")
else:
    print("You must be 18+")
```

If you only need one yes/no answer, this may be simpler:

```
if age >= 18 and has_ticket:
```



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22. Input Validation with Selection

```
mark = int(input("Enter mark (0-100): "))

if mark < 0 or mark > 100:
    print("Invalid mark")
else:
    print("Valid mark")
```



22. Input Validation with Selection

```
mark = int(input("Enter mark (0-100): "))

if mark < 0 or mark > 100:
    print("Invalid mark")
else:
    print("Valid mark")
```

Selection can detect bad input — but it does not ask again.

Iteration will later let us repeat until the input is valid.



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23. Full Example: Mark Report

```
mark = int(input("Enter mark: "))

if mark < 0 or mark > 100:
    print("Invalid mark")
elif mark >= 70:
    print("Distinction")
elif mark >= 40:
    print("Pass")
else:
    print("Fail")
```



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24. Full Example: Delivery Charge

```
order = float(input("Order value: €"))
member = input("Member? y/n: ").lower()

if order >= 50 or member == "y":
    delivery = 0
else:
    delivery = 5

print("Delivery: €", delivery)
```





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25. Debugging Challenge

Find the problems:

```
age = int(input("Age: "))

if age >= 13 and <= 19:
    print("Teenager")
elif age > 19
    print("Adult")
else:
    print("Child")
```



25. Debugging Challenge

Find the problems:

```
age = int(input("Age: "))

if age >= 13 and <= 19:
    print("Teenager")
elif age > 19
    print("Adult")
else:
    print("Child")
```

There are three syntax/structure problems.



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26. Debugging Solution

```
age = int(input("Age: "))

if 13 <= age <= 19:
    print("Teenager")
elif age > 19:
    print("Adult")
else:
    print("Child")
```



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27. Design Activity: Cinema Ticket

Design a program using these rules:

- Under 5 $\rightarrow$ Free
- 5–15 $\rightarrow$ €8
- 16–64 $\rightarrow$ €12
- 65+ $\rightarrow$ €9

Before coding, write the conditions in the order Python should test them.





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28. Cinema Ticket Solution

```
age = int(input("Age: "))

if age < 5:
    price = 0
elif age <= 15:
    price = 8
elif age <= 64:
    price = 12
else:
    price = 9

print("Ticket price: €", price)
```



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29. Mini Challenge: Leap Year Thinking

A simplified rule:

A year is a leap year if it is divisible by 4.

Hint: modulo % gives the remainder.

```
year % 4 == 0
```



29. Mini Challenge: Leap Year Thinking

A simplified rule:

A year is a leap year if it is divisible by 4.

Hint: modulo % gives the remainder.

`year % 4 == 0`

Write a program that prints Leap year or Not a leap year.



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30. Leap Year — Simplified Solution

The simplified rule:

A year is a leap year if it is divisible by 4.

```
year = int(input("Enter a year: "))
```

```
if year % 4 == 0:  
    print("Leap year")  
else:  
    print("Not a leap year")
```



30. Leap Year — Simplified Solution

The simplified rule:

A year is a leap year if it is divisible by 4.

```
year = int(input("Enter a year: "))
```

```
if year % 4 == 0:
    print("Leap year")
else:
    print("Not a leap year")
```

This is a useful first solution for practising modulo % and if-else, but it does not cover the full leap-year rule.



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31. Leap Year — Full Solution

The full rule also needs to account for century years:

```
year = int(input("Enter a year: "))

if year % 4 == 0:
    if year % 100 == 0:
        if year % 400 == 0:
            print("Leap year")
        else:
            print("Not a leap year")
    else:
        print("Leap year")
else:
    print("Not a leap year")
```



31. Leap Year — Full Solution

The full rule also needs to account for century years:

```
year = int(input("Enter a year: "))

if year % 4 == 0:
    if year % 100 == 0:
        if year % 400 == 0:
            print("Leap year")
        else:
            print("Not a leap year")
    else:
        print("Leap year")
else:
    print("Not a leap year")
```

For example: 2000 is a leap year, but 1900 is not.



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32. Final Check

Which structure would you choose?

1. Print a warning if temperature is below zero.
2. Print Pass or Fail.
3. Classify a mark as Distinction / Pass / Fail.
4. Allow entry only if age and ticket conditions are satisfied.



32. Final Check

Which structure would you choose?

1. Print a warning if temperature is below zero.
2. Print Pass or Fail.
3. Classify a mark as Distinction / Pass / Fail.
4. Allow entry only if age and ticket conditions are satisfied.

`if`, `if-else`, `if-elif-else`, and.



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33. Session 2 Recap

You should now be able to use:

- `if, elif, else`
- separate versus mutually-exclusive conditions
- `and, or, not`
- chained comparisons
- nested selection
- selection for basic validation

Most importantly: **write the decision rules clearly before writing the Python.**